MODULE 1 – MATHEMATICAL LOGIC

Rules of Inference, Natural Deduction, Methods of Proof and Resolution Principle

1 Rules of Inference

1.1 Introduction

In mathematical logic, we often know some statements to be true and want to use them to obtain another statement.

For example:

> If a student studies, then the student will pass.
> Anu studies.
> Therefore, Anu will pass.

The first two statements are given information, and the third statement is obtained by logical reasoning.

The process of obtaining a conclusion from given statements is called **inference**.

1.2 Argument

Definition

An **argument** is a collection of statements in which some statements are given as reasons for accepting another statement.

An argument consists of:

1. Premises
2. Conclusion

1.2.1 Premise

A **premise** is a statement that is given or assumed to be true.

1.2.2 Conclusion

A **conclusion** is the statement that is obtained from the premises using logical reasoning.

Example

Consider:

> If it rains, the ground is wet.
> It is raining.
> Therefore, the ground is wet.

Let

$$p = \text{It is raining}$$

and

$$q = \text{The ground is wet}$$

Then the argument can be written as:

$$p \to q$$

$$p$$

$$\therefore q$$

Here,

- $p \to q$ is a premise.
- p is a premise.
- q is the conclusion.

1.3 Rule of Inference

Definition

A **rule of inference** is a standard and valid pattern of reasoning used to obtain a conclusion from one or more premises.

In simple words:

> A rule of inference tells us **how we can logically move from known statements to a new statement**.

The general form is:

$$\frac{\text{Premises}}{\therefore \text{Conclusion}}$$

The symbol $\therefore$ means "therefore."

1.4 Important Rules of Inference

The important rules of inference are:

1. Modus Ponens
2. Modus Tollens
3. Hypothetical Syllogism
4. Disjunctive Syllogism
5. Addition
6. Simplification
7. Conjunction
8. Resolution
9. Constructive Dilemma
10. Destructive Dilemma

The first seven are especially important for writing the standard rules of inference.

1.5 Modus Ponens

Definition

Modus Ponens is a rule of inference in which the antecedent of an implication is affirmed, allowing us to conclude its consequent.

The general form is:

$$p \to q$$

$$p$$

$$\therefore q$$

If p implies q, and p is true, then q must be true.

Example

Example:
If I study, then I will pass.
I study.
Therefore, I will pass.
Let

$$p = \text{I study}$$

$$q = \text{I will pass}$$

Then:

$$p \to q$$

$$p$$

Therefore:

$$q$$

Hence, the argument uses **Modus Ponens**.

Important

Remember:

$$\boxed{p \to q, \quad p \quad \therefore q}$$

Confirm p, conclude q.

1.6 Modus Tollens

Definition

Modus Tollens is a rule of inference in which the consequent of an implication is denied, allowing us to conclude that the antecedent is false.

The general form is:

$$p \to q$$

$$\neg q$$

$$\therefore \neg p$$

Example

Example:
If it rains, the ground is wet.
The ground is not wet.
Therefore, it did not rain.
Let

$$p = \text{It rains}$$

$$q = \text{The ground is wet}$$

Then:

$$p \to q$$

$$\neg q$$

Therefore:

$$\neg p$$

Hence, this is **Modus Tollens**.

Important

Remember:

$$\boxed{p \to q, \quad \neg q \quad \therefore \neg p}$$

Deny q, conclude $\neg p$.

1.7 Hypothetical Syllogism

The general form is:

$$p \to q$$

$$q \to r$$

$$\therefore p \to r$$

If p leads to q, and q leads to r, then p leads to r.

Example

If I study, I understand the subject.
If I understand the subject, I get good marks.
Therefore, if I study, I get good marks.
Symbolically:

$$p \to q$$

$$q \to r$$

Therefore:

$$p \to r$$

This is called **Hypothetical Syllogism**.

Important

The middle proposition connects the two implications:

$$p \to \boxed{q}$$

$$\boxed{q} \to r$$

Therefore:

$$p \to r$$

1.8 Disjunctive Syllogism

The general form is:

$$p \vee q$$

$$\neg p$$

$$\therefore q$$

If p or q is true and p is false, then q must be true.

Example

I will study Mathematics or Programming.
I will not study Mathematics.
Therefore, I will study Programming.
Let:

$$p = \text{I will study Mathematics}$$

$$q = \text{I will study Programming}$$

Then:

$$p \vee q$$

$$\neg p$$

Therefore:

$$q$$

Hence, the rule is **Disjunctive Syllogism**.

Another valid form is:

$$p \vee q, \quad \neg q \quad \therefore p$$

1.9 Addition

The general form is:

$$p$$

$$\therefore p \vee q$$

If p is true, then $p \vee q$ is true, regardless of whether q is true or false.

Example

Given:

It is raining.

Therefore:

It is raining or the sun is shining.

Symbolically:

$$p$$

Therefore:

$$p \vee q$$

This rule is called **Addition**.

1.10 Simplification

The general form is:

$$p \wedge q$$

$$\therefore p$$

or

$$p \wedge q$$

$$\therefore q$$

If a conjunction $p \wedge q$ is true, then both p and q are true individually.

Example

Given:

Ravi is a student and Ravi is hardworking.

Let:

$$p = \text{Ravi is a student}$$

$$q = \text{Ravi is hardworking}$$

Then:

$$p \wedge q$$

Therefore, we can conclude:

$$p$$

We can also conclude:

$$q$$

This rule is called **Simplification**.

Important

Simplification means **splitting an AND statement**.

$$\boxed{p \wedge q \Rightarrow p}$$

or

$$\boxed{p \wedge q \Rightarrow q}$$

1.11 Conjunction

The general form is:

$$p$$

$$q$$

$$\therefore p \wedge q$$

If p is true and q is true, then they can be combined using the AND connective.

Example

Given:

Ravi is a student.

and

Ravi is hardworking.

Therefore:

$$p \wedge q$$

Thus:

Ravi is a student and Ravi is hardworking.

This rule is called **Conjunction**.

1.12 Conjunction and Simplification

These two rules are opposite in nature.

Rule	Conjunction	Simplification
Meaning	Combines two statements	Separates a conjunction
Form	$p,\ q \Rightarrow p \wedge q$	$p \wedge q \Rightarrow p$ or q
Idea	JOIN	SPLIT

Therefore, remember:

$$\boxed{p, q \Rightarrow p \wedge q}$$

is **Conjunction**.
and

$$\boxed{p \wedge q \Rightarrow p \text{ or } q}$$

is **Simplification**.

1.13 Summary of Important Rules of Inference

Rule	Logical Form
Modus Ponens	$p \to q,\ p \therefore q$
Modus Tollens	$p \to q,\ \neg q \therefore \neg p$
Hypothetical Syllogism	$p \to q,\ q \to r \therefore p \to r$
Disjunctive Syllogism	$p \vee q,\ \neg p \therefore q$
Addition	$p \therefore p \vee q$
Simplification	$p \wedge q \therefore p$
Conjunction	$p, q \therefore p \wedge q$

2 Natural Deduction

2.1 Introduction

Rules of inference provide individual patterns of valid reasoning. However, a complicated argument may require several rules to be applied one after another.

Natural deduction provides a systematic way of doing this.

Definition

Natural deduction is a formal method of proving a conclusion from given premises by applying valid rules of inference step by step.

In simple words:

> Natural deduction is a method in which we start with the given premises and derive the required conclusion one step at a time, giving a reason for every step.

2.2 General Structure of Natural Deduction

For example, suppose:

$$p \to q$$

$$p$$

We want to prove:

$$q$$

The natural deduction is:

1.	$p \to q$	Premise
2.	p	Premise
3.	q	Modus Ponens from (1),(2)

Every statement has a logical justification.

2.3 Importance of Natural Deduction

Natural deduction helps us to:

- prove arguments systematically,
- derive conclusions step by step,
- identify the exact reason for each step,
- avoid unsupported conclusions,
- solve logical proof problems.

2.4 Introduction and Elimination Rules

In natural deduction, rules are often classified as **introduction rules** and **elimination rules**.

2.4.1 Introduction Rule

An introduction rule is used to introduce or construct a logical connective.

2.4.2 Elimination Rule

An elimination rule is used to obtain a simpler statement from a statement containing a logical connective.

2.5 Conjunction Introduction

If we know:

$$p$$

and:

$$q$$

we can introduce the conjunction:

$$p \wedge q$$

Therefore:

$$\boxed{\frac{p \qquad q}{p \wedge q}}$$

This is also called **AND Introduction**.

2.6 Conjunction Elimination

If we know:

$$p \wedge q$$

we can obtain either:

$$p$$

or:

$$q$$

Therefore:

$$\boxed{\frac{p \wedge q}{p}}$$

and

$$\boxed{\frac{p \wedge q}{q}}$$

These are called **AND Elimination** rules.

2.7 Implication Elimination

From:

$$p \rightarrow q$$

and:

$$p$$

we can derive:

$$q$$

Therefore:

$$\boxed{\frac{p \to q \qquad p}{q}}$$

This is Modus Ponens and is also called **Implication Elimination**.

2.8 Implication Introduction

Suppose we want to prove:

$$p \to q$$

A common approach is:

1. Assume p.
2. Derive q.
3. Therefore conclude $p \to q$.

The basic idea is:

$$\boxed{\text{Assume } p \to \text{derive } q \to p \to q}$$

This is closely related to the **direct method of proof**.

2.9 Disjunction Introduction

If p is true, then:

$$p \vee q$$

can be concluded.
Therefore:

$$\boxed{\frac{p}{p \vee q}}$$

Similarly, if q is true:

$$\boxed{\frac{q}{p \vee q}}$$

2.10 Negation and Contradiction

Negation is represented by:

$$\neg p$$

To prove:

$$\neg p$$

we can assume:

$$p$$

and show that this assumption leads to a contradiction.
Therefore:

$$\boxed{\text{Assume } p \to \text{contradiction} \to \neg p}$$

This idea forms the basis of **proof by contradiction**.

2.11 Example of Natural Deduction

Example

Prove:

$$(p \wedge q) \to p$$

Solution:
We want to prove an implication, so assume:

$$p \wedge q$$

From $p \wedge q$, using AND Elimination:

$$p$$

Therefore:

$$(p \wedge q) \to p$$

In tabular form:

1.	$p \wedge q$	Assumption
2.	p	AND Elimination from (1)
3.	$(p \wedge q) \to p$	Implication Introduction

3 Methods of Proof

3.1 Introduction

A **proof** is a logical argument used to establish that a mathematical statement is true.

Different statements may require different approaches. Therefore, several methods of proof are used in mathematics.

The important methods are:

1. Direct proof
2. Proof by contrapositive
3. Proof by contradiction
4. Proof by cases
5. Proof by exhaustion
6. Proof by counterexample

3.2 Direct Proof

Definition

In a **direct proof**, we start with the given assumption and use logical or mathematical steps to derive the required conclusion.

General structure:

$$\boxed{\text{Given} \rightarrow \text{Logical steps} \rightarrow \text{Conclusion}}$$

Example

Prove:
If n is even, then n^2 is even.
Step 1: Start with the assumption
Since n is even:

$$n = 2k$$

where k is an integer.
Step 2: Square both sides

$$n^2 = (2k)^2$$

$$n^2 = 4k^2$$

Step 3: Write in the form $2m$

$$n^2 = 2(2k^2)$$

Since $2k^2$ is an integer, n^2 is divisible by 2.
Therefore:

$$\boxed{n^2 \text{ is even}}$$

Hence proved.

3.3 Proof by Contrapositive

Suppose we want to prove:

$$p \to q$$

The contrapositive is:

$$\neg q \to \neg p$$

A proposition and its contrapositive are logically equivalent:

$$\boxed{p \to q \equiv \neg q \to \neg p}$$

Therefore, instead of proving $p \to q$ directly, we can prove:

$$\neg q \to \neg p$$

Example

Statement:
If n^2 is even, then n is even.
Contrapositive:
If n is not even, then n^2 is not even.
If n is not even, then n is odd.
Therefore:

$$n = 2k + 1$$

Now:

$$n^2 = (2k + 1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

Therefore, n^2 is odd.
So n^2 is not even.
Hence the contrapositive is true, and therefore:

$$\boxed{n^2 \text{ even} \rightarrow n \text{ even}}$$

is proved.

3.4 Proof by Contradiction

Definition

In proof by contradiction, we assume that the statement to be proved is false and show that this assumption leads to a contradiction.

General structure:

$$\boxed{\text{Assume opposite} \rightarrow \text{Contradiction} \rightarrow \text{Original statement is true}}$$

Example

Prove:

$$\sqrt{2}$$

is irrational.

Step 1: Assume the opposite

Assume $\sqrt{2}$ is rational.

Then:

$$\sqrt{2} = \frac{a}{b}$$

where a and b are integers, $b \neq 0$, and a/b is in lowest terms.

Step 2: Square both sides

$$2 = \frac{a^2}{b^2}$$

Therefore:

$$a^2 = 2b^2$$

So a^2 is even.

Hence a is even.

Let:

$$a = 2k$$

Step 3: Substitute

$$(2k)^2 = 2b^2$$

$$4k^2 = 2b^2$$

Therefore:

$$b^2 = 2k^2$$

So b^2 is even.

Hence b is also even.

Therefore both a and b are even.

But this contradicts the assumption that a/b is in lowest terms.

Therefore our original assumption is false.

Hence:

$$\boxed{\sqrt{2} \text{ is irrational}}$$

3.5 Proof by Cases

Definition

In proof by cases, a problem is divided into all possible cases, and the statement is proved separately for each case.

For example, every integer is either:

- even, or
- odd.

Suppose we want to prove a property for every integer n.
Case 1: n is even.
Prove that the required property is true.
Case 2: n is odd.
Prove that the required property is true.
Since every integer is either even or odd, the property is true for every integer.

3.6 Proof by Exhaustion

Definition

Proof by exhaustion is a method in which every possible case in a finite set is checked.

Example

Suppose a statement needs to be verified for:

$$1, 2, 3, 4, 5$$

We check the statement separately for each value.
If it is true for all five values, then it has been proved for this finite set.

3.7 Proof by Counterexample

A **counterexample** is a particular example that shows a universal statement is false.

Example

Consider the statement:

Every prime number is odd.

Take:

2

The number 2 is prime, but 2 is even.
Therefore the statement is false.
Thus:

2 is a counterexample

Important

To prove a universal statement, one must generally establish it for all relevant cases. But to **disprove** a universal statement, only **one counterexample** is sufficient.

3.8 Summary of Methods of Proof

Method	Main idea
Direct proof	Start from the assumption and derive the conclusion.
Contrapositive	Prove $\neg q \to \neg p$ instead of $p \to q$.
Contradiction	Assume the opposite and derive a contradiction.
Cases	Divide the problem into cases and prove each.
Exhaustion	Check every possibility in a finite set.
Counterexample	Find one example that disproves a universal statement.

4 Resolution Principle

4.1 Introduction

The **Resolution Principle** is an important rule of inference used in mathematical logic and automated reasoning.

It is particularly useful in:

- theorem proving,
- artificial intelligence,
- automated reasoning,
- logic programming,
- knowledge representation.

4.2 Literal

Definition

A **literal** is either a proposition or its negation.

Examples:

$$p, \quad q, \quad r$$

and

$$\neg p, \quad \neg q, \quad \neg r$$

are all literals.
Thus:

$$\boxed{p \text{ and } \neg p \text{ are both literals}}$$

4.3 Complementary Literals

Definition

Two literals are called **complementary literals** if one is the negation of the other.

Examples:

$$p, \neg p$$

$$q, \neg q$$

$$r, \neg r$$

In the resolution process, complementary literals are eliminated.

4.4 Clause

Definition

A **clause** is a disjunction (OR) of literals.

Examples:

$$p \vee q$$

$$\neg p \vee r$$

$$p \vee \neg q \vee r$$

are clauses.

4.5 Resolution Rule

The basic resolution rule is:

$$p \vee q$$

$$\neg p \vee r$$

Therefore:

$$\boxed{q \vee r}$$

It can be written as:

$$\boxed{\frac{p \vee q, \quad \neg p \vee r}{q \vee r}}$$

Here:

$$p, \neg p$$

are complementary literals.

4.6 How Resolution Works

Consider:

$$p \vee q$$

and:

$$\neg p \vee r$$

Step 1: Identify complementary literals

$$p, \neg p$$

Step 2: Eliminate the complementary pair
Remove p and $\neg p$.
Step 3: Combine the remaining literals
The remaining literals are:

$$q, r$$

Therefore:

$$\boxed{q \vee r}$$

4.7 Example

Example

Resolve:

$$p \vee q$$

$$\neg p \vee r$$

Solution:
Complementary pair:

$$p, \neg p$$

Remove them.
Remaining literals:

$$q, r$$

Therefore:

$$\boxed{q \vee r}$$

4.8 Another Example

Example

Resolve:

$$p \vee q$$

$$\neg q \vee r$$

The complementary literals are:

$$q, \neg q$$

After removing them, the remaining literals are:

$$p, r$$

Therefore:

$$\boxed{p \vee r}$$

4.9 Conjunctive Normal Form

Resolution is normally applied to clauses, so expressions are commonly converted into **Conjunctive Normal Form (CNF)**.

Definition

A logical expression is in **Conjunctive Normal Form** when it is an AND of clauses, where each clause is an OR of literals.

General form:

$$(C_1) \wedge (C_2) \wedge \cdots \wedge (C_n)$$

where every C_i is a disjunction of literals.

Example

The expression

$$(p \vee q) \wedge (\neg p \vee r)$$

is in CNF.
It contains two clauses:

$$C_1 = p \vee q$$

$$C_2 = \neg p \vee r$$

4.10 Resolution Method for Proving Validity

Resolution can be used to determine whether an argument is valid.

Suppose the argument is:

$$P_1, P_2, \ldots, P_n \therefore Q$$

The general procedure is:

1. Write the premises.
2. Negate the conclusion.
3. Combine all statements.
4. Convert the complete expression into CNF.
5. Separate the CNF expression into clauses.
6. Apply the resolution rule.
7. Continue until a conclusion is reached.

If the **empty clause** is obtained:

$$\boxed{\square}$$

then a contradiction has been obtained.
Therefore, the original argument is valid.

4.11 Empty Clause

The empty clause is represented by:

$$\boxed{\square}$$

It means that no literal remains and a contradiction has been obtained.
Therefore:

$$\boxed{\text{Derivation of } \square \Rightarrow \text{Contradiction}}$$

and hence:

$$\boxed{\text{Contradiction} \Rightarrow \text{Original argument is valid}}$$

4.12 Example – Proving Validity Using Resolution

Consider:

$$p \rightarrow q$$

$$p$$

Therefore:

$$q$$

We will prove this using resolution.

Step 1: Negate the conclusion

The conclusion is:

$$q$$

Therefore, assume:

$$\neg q$$

Step 2: Convert the implication

We know:

$$p \rightarrow q \equiv \neg p \vee q$$

Therefore, the clauses are:

$$\neg p \vee q$$

$$p$$

$$\neg q$$

Step 3: Resolve

Resolve:

$$\neg p \vee q$$

with:

$$\neg q$$

The complementary literals are:

$$q, \neg q$$

Therefore:

$$\neg p$$

Step 4: Resolve again

We now have:

$$p$$

and:

$$\neg p$$

These are complementary.
Therefore:

$$\boxed{\square}$$

An empty clause is obtained.
A contradiction has been obtained.
Therefore, the original argument is valid:

$$\boxed{p \rightarrow q, \quad p \therefore q}$$

5 Difference Between the Four Topics

Students should not confuse these concepts.

Topic	Meaning
Rules of Inference	Standard patterns used to derive conclusions.
Natural Deduction	Step-by-step deduction using logical rules.
Methods of Proof	Different strategies for proving mathematical statements.
Resolution Principle	A specific inference technique using clauses and complementary literals.

In Simple Words

Rules of inference are the individual logical rules.

Natural deduction uses these rules step by step to construct a proof.

Methods of proof are different approaches such as direct proof, contrapositive, and contradiction.

Resolution is a particular systematic technique that works with clauses and complementary literals.

6 Important Exam-Oriented Questions

6.1 Short-Answer Questions

1. Define an argument.
2. What is a premise?
3. What is a conclusion?
4. Define a rule of inference.
5. State Modus Ponens.
6. State Modus Tollens.
7. State Hypothetical Syllogism.
8. State Disjunctive Syllogism.
9. Define natural deduction.
10. What is a direct proof?

11. What is proof by contradiction?
12. What is proof by contrapositive?
13. Define a literal.
14. What are complementary literals?
15. Define a clause.
16. What is CNF?
17. State the resolution principle.
18. What is an empty clause?

6.2 3-Mark Questions

1. Write any three rules of inference.
2. Explain Modus Ponens with an example.
3. Explain Modus Tollens with an example.
4. Explain proof by contradiction.
5. Explain proof by contrapositive.
6. Define literal and complementary literals.
7. Explain the resolution principle with an example.

6.3 7-Mark Questions

1. List any seven rules of inference with their logical forms.
2. Establish the validity of a given argument using rules of inference.
3. Explain the different methods of proof with examples.
4. Prove a given statement using direct proof.
5. Prove a given statement using proof by contradiction.
6. Prove a given statement using the contrapositive method.
7. Explain the resolution principle with suitable examples.
8. Establish the validity of an argument using the resolution principle.

7 Quick Revision

Rules of Inference

$$\boxed{p \to q,\ p \Rightarrow q}$$

Modus Ponens

$$\boxed{p \to q,\ \neg q \Rightarrow \neg p}$$

Modus Tollens

$$\boxed{p \to q,\ q \to r \Rightarrow p \to r}$$

Hypothetical Syllogism

$$\boxed{p \vee q,\ \neg p \Rightarrow q}$$

Disjunctive Syllogism

$$\boxed{p \Rightarrow p \vee q}$$

Addition

$$\boxed{p \wedge q \Rightarrow p \text{ or } q}$$

Simplification

$$\boxed{p, q \Rightarrow p \wedge q}$$

Conjunction

Natural Deduction

$$\boxed{\text{Premises} \to \text{Logical steps} \to \text{Conclusion}}$$

Every step must have a valid reason.

Methods of Proof

$$\boxed{\text{Direct proof}}$$

$$\boxed{\text{Contrapositive}}$$

$$\boxed{\text{Contradiction}}$$

Cases

Exhaustion

Counterexample

Resolution

Basic resolution rule:

$$(p \lor q), (\neg p \lor r) \Rightarrow (q \lor r)$$

Remember:

Find complementary literals → eliminate them → combine remaining literals

For proving validity:

Negate conclusion → Convert to CNF → Form clauses → Apply resolution → $\square$

If the empty clause $\square$ is obtained, a contradiction has been reached, and the original argument is valid.